Models aren’t magic:
They’re statistics at scale.

ChatGPT 4o mini

How many r's are there in strawberry?

There are two **r's** in the word "strawberry."

BER
MUERJIE
MARJA
VOTES
BÆR
BĀR
CAOR
BERRY
BERRIE
BERIE
BERY
MOREN
BEI
BEIER
BEERE
BOUBLE
JÁHADA
JÁGODA
JÁHODA
BAIE
BERRI
PER
BOGYÓ
JAGODA
BACĂ
BAIA
BAGA
BAYA
BACCA
BOBICA
BACA
BAIKA
ARCTIC OCEAN
Arctic Circle
NORWEGIAN SEA
ICELAND
NORWAY
SWEDEN
FINLAND
Helsinki
Oslo
Tallinn
ESTONIA
RUSSIA
ASIA
UNITED KINGDOM
SCOTLAND
N. IRELAND
IRELAND
WALES
NORTH SEA
DENMARK
Copenhagen
BALTIC SEA
Riga
LATVIA
Moscow
LITHUANIA
Vilnius
ATLANTIC OCEAN
POLAND
Warsaw
KAZAKHSTAN
Berlin
Paris
Luxembourg
EUROPE
Kyiv
Prague
Vienna
SLVK.
MOLDOVA
Chisinau
SWITZ.
HUNGARY
SLVN.
Ljubljana
CASPIAN SEA
PORTUGAL
ANDORRA
Andorra la Vella
MONACO
Monaco
San Marino
CRO.
BOS. HER.
Belgrade
Bucharest
BLACK SEA
SPAIN
Sarajevo
Sofia
BULGARIA
VATICAN CITY
Podgorica
KOS.
Pristina
Scopje
TURKEY
Tirana
N. MACED.
ALB.
ITALY
MEDITERRANEAN SEA
GREECE
Athens
AFRICA
MALTA
Valletta
CYPRUS
Nicosia
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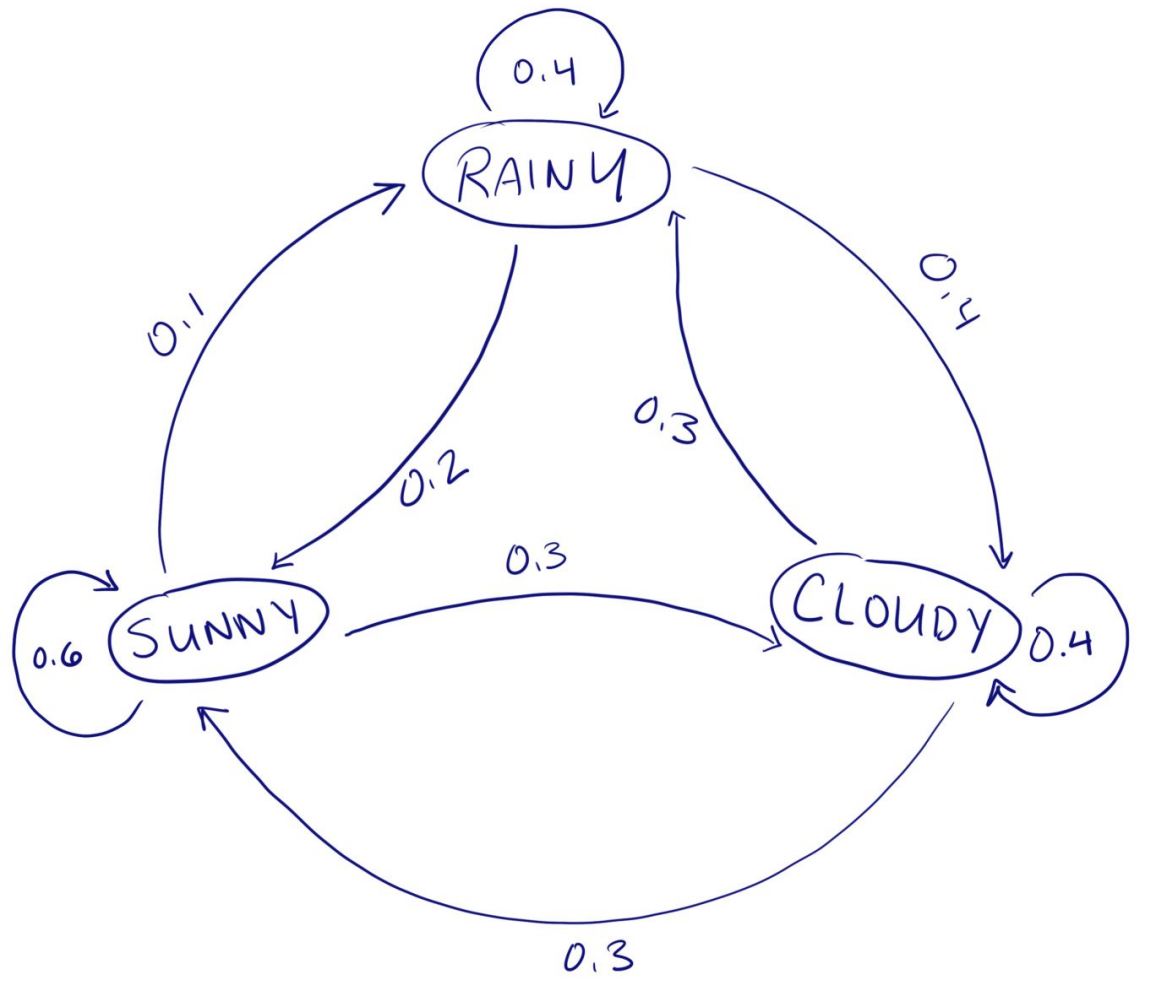
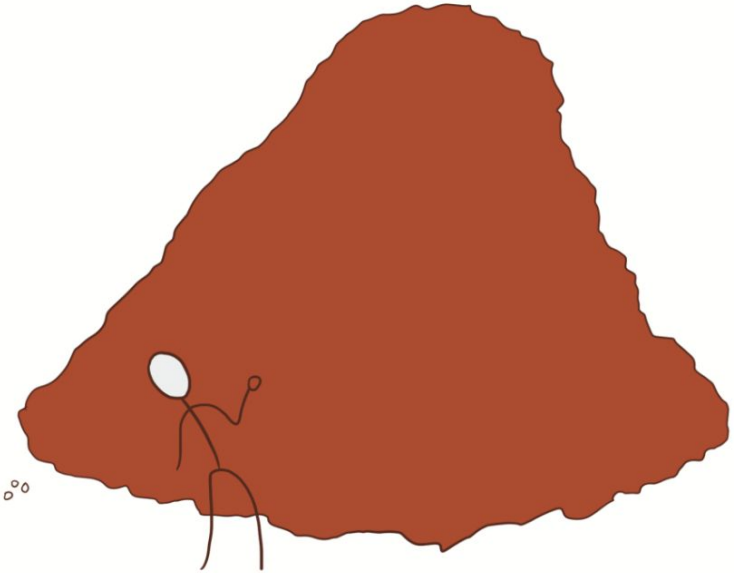
0 150 300 mi
0 200 400 km

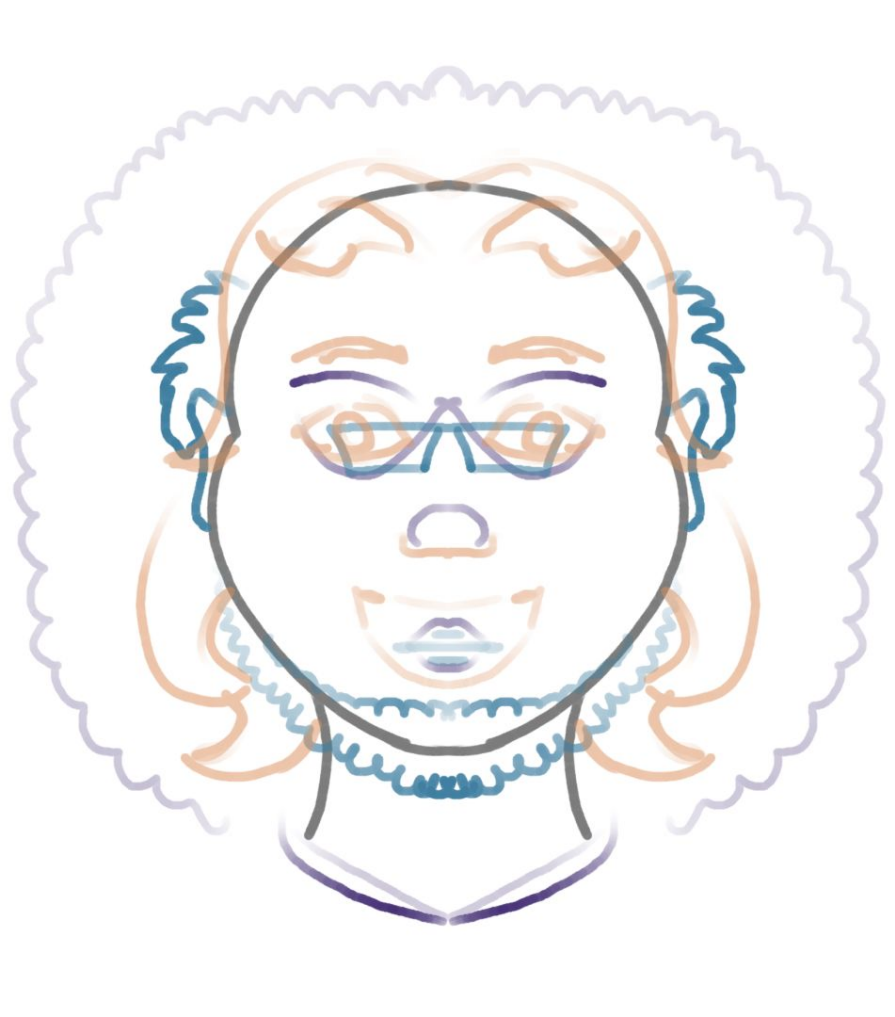

0.4
RAINY
0.1
0.2
0.3
0.4
0.3
0.6
SUNNY
CLOUDY
0.4
0.3

| | | | | | |
|---|---|---|---|---|---|
| **Input Parameter Name:** | | | | | |
| n - 6 | | | | | |
| n - 5 | **[START]** | | | | |
| n - 4 | I | | | | |
| n - 3 | went | | | | |
| n - 2 | to | | | | |
| n - 1 | the | | | | |
| **prediction for n:** | store | movies | car | apple | parrot |
| **weight of prediction:** | 0.04 | 0.04 | 0.03 | 0.0001 | 0.0006 |

| | | | | | |
|---|---|---|---|---|---|
| **Input Parameter Name:** | | | | | |
| n - 6 | | | | | |
| n - 5 | **[START]** | | | | |
| n - 4 | I | | | | |
| n - 3 | went | | | | |
| n - 2 | to | | | | |
| n - 1 | the | | | | |
| **prediction for n:** | store | movies | car | apple | parrot |
| **weight of prediction:** | 0.04 | 0.04 | 0.03 | 0.0001 | 0.0006 |

| | | | | | |
|---|---|---|---|---|---|
| **Input Parameter Name:** | | | | | |
| n - 6 | | **[START]** | | | |
| n - 5 | **[START]** | I | | | |
| n - 4 | I | went | | | |
| n - 3 | went | to | | | |
| n - 2 | to | the | | | |
| n - 1 | the | store | | | |
| **prediction for n:** | | to | and | apple | parrot |
| **weight of prediction:** | | 0.09 | 0.04 | 0.00001 | 0.00002 |





| | | | | | |
|---|---|---|---|---|---|
| **Input Parameter Name:** | | | | | |
| n - 6 | | | | | |
| n - 5 | Justin | | | | |
| n - 4 | Bieber | | | | |
| n - 3 | was | | | | |
| n - 2 | born | | | | |
| n - 1 | in | | | | |
| **prediction for n:** | 1994 | 1900 | 2002 | apple | parrot |
| **weight of prediction:** | 0.8 | 0.02 | 0.07 | 0.00001 | 0.00002 |

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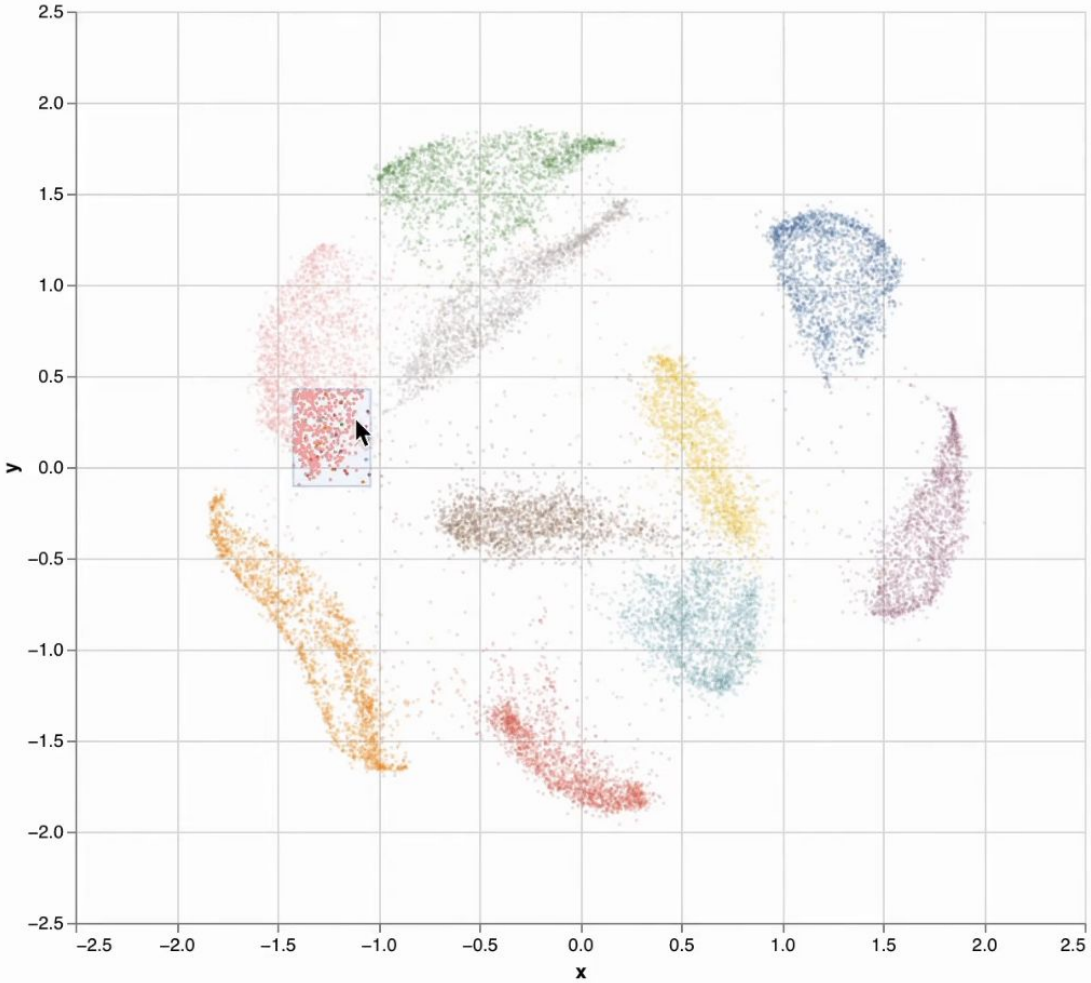
https://johnsonjohnson.law › orland-park

Orland Park Law Office

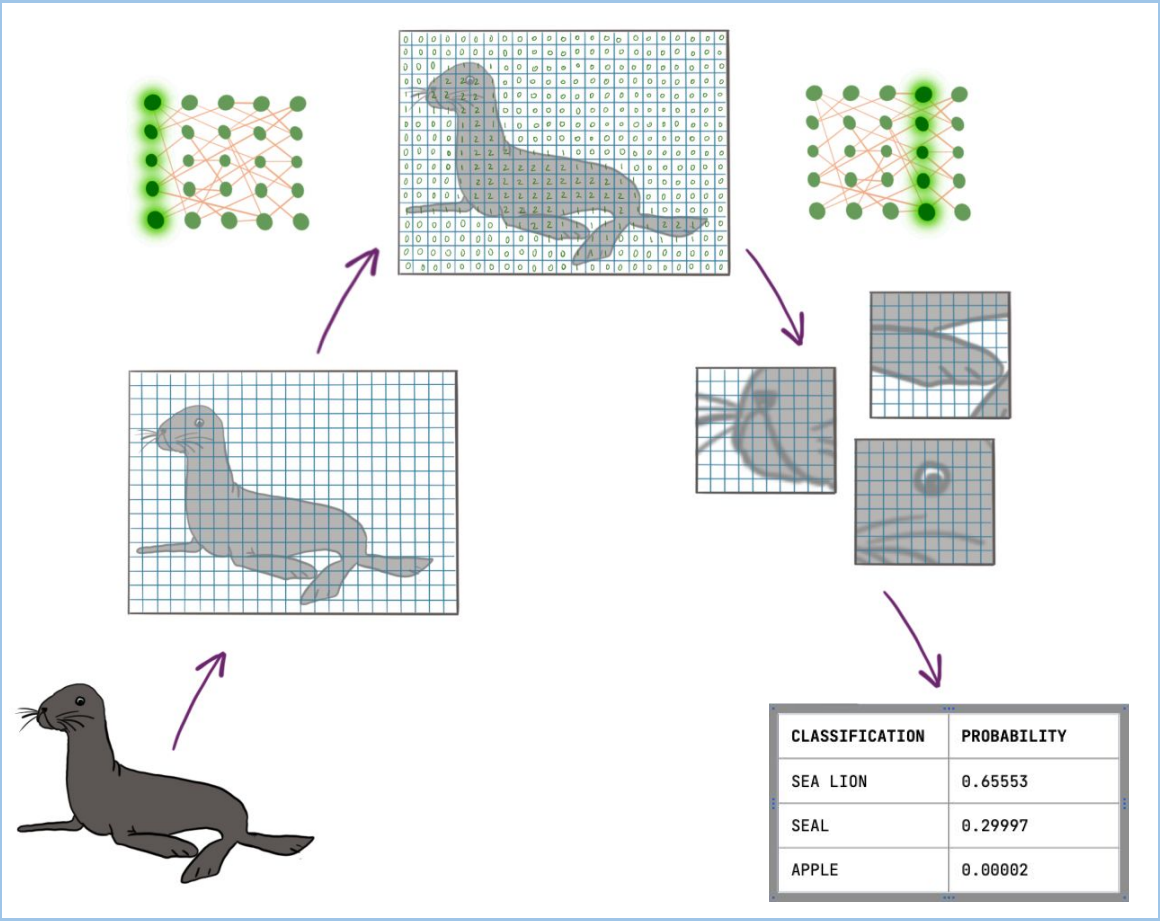
**Johnson & Johnson, Ltd.** has an Orland Park law office located in Orland Park, Illinois, on the northeast corner of 167th Street and 108th Avenue.

| **Input Parameter Name:** | | | | | |
|---|---|---|---|---|---|
| n - 6 | | | | | |
| n - 5 | The | | | | |
| n - 4 | law | | | | |
| n - 3 | firm | | | | |
| n - 2 | Johnson | | | | |
| n - 1 | and | | | | |
| **prediction for n:** | Bell | Associates | Stone | Sullivan | Murray |
| **weight of prediction:** | 0.1 | 0.1 | 0.1 | 0.1 | 0.1 |




| | |
|---|---|
| acute | learned |
| astute | nimble |
| brilliant | perspicacious |
| cerebral | quick-witted |
| cunning | resourceful |
| discerning | sagacious |
| erudite | sapient |
| ingenious | savvy |
| intelligent | sharp-witted |
| judicious | shrewd |
| keen | wise |

# “Thinking” Implementation:



| CLASSIFICATION | PROBABILITY |
|---|---|
| SEA LION | 0.65553 |
| SEAL | 0.29997 |
| APPLE | 0.00002 |

# What characteristics of code help to explain the success of LLMs like Opus?

# What characteristics of code help to explain the success of LLMs like Opus?

1. Limited Vocabulary

# What characteristics of code help to explain the success of LLMs like Opus?

1. Limited Vocabulary
2. Limited and strictly enforced grammar

# What characteristics of code help to explain the success of LLMs like Opus?

1. Limited Vocabulary
2. Limited and strictly enforced grammar
3. Centralized change procedure

# What characteristics of code help to explain the success of LLMs like Opus?

1. Limited Vocabulary
2. Limited and strictly enforced grammar
3. Centralized change procedure
4. Modular libraries for solution composition

# What characteristics of code help to explain the success of LLMs like Opus?

1. Limited Vocabulary
2. Limited and strictly enforced grammar
3. Centralized change procedure
4. Modular libraries for solution composition
5. Somewhat strong case for customization